

Archit Dewan

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EDUCATION

McMaster University

Honours BSc in Computer Science and Mathematics

Sep. 2024 – May 2028

Hamilton, Ontario

Relevant Coursework: Data Structures and Algorithms, Software Development, Functional Programming, Computer Architecture, Discrete Mathematics

OPEN SOURCE

Traefik | *Open Source Contributor* | *65k GitHub Stars*

[PRs](#)

- Implemented adaptive sticky-session cookie handling for Traefik's Kubernetes ingress-nginx provider across three load-balancing strategies, preserving session affinity for legacy browsers through comprehensive compatibility testing.
- Fixed ingress-nginx **server-snippet** path matching by correctly handling the NGINX prefix modifier, adding regression tests for matching and non-matching request paths and documenting supported behavior.

PROJECTS

Incident Response | *Go, PostgreSQL, Python, Docker, OpenAI*

[Repo](#) 2026

- Built a Go and PostgreSQL incident-response tool that organizes simulated outage alerts into tracked incidents, **grouping related alerts** and preserving a timestamped response history.
- Centralized outage context in **Slack** by matching alerts to troubleshooting runbooks and attaching severity and simulated customer-impact summaries through webhook notifications.
- Automated **post-incident report drafting** with a separate Python/OpenAI worker that combines alerts, response timelines, impact snapshots, and runbook metadata, keeping AI generation outside the Go API's request path.

API Monitoring Dashboard | *Go, PostgreSQL, Docker, Oracle Cloud*

[Repo](#) 2026

- Automated recurring **API availability checks** with a Go scheduler, using configurable HTTP status expectations and timeouts to identify failed requests without manually testing each endpoint.
- Made recurring failures and slow responses visible through a **PostgreSQL-backed dashboard** displaying consecutive failures, current incident start times, and historical latency charts.
- Automated releases to an **Oracle Cloud VM** using GitHub Actions, SSH, and Docker Compose, serving the dashboard over HTTPS through Caddy and checking public endpoint availability after deployment.

Surge Prep | *Hack the North* | *Python, FastAPI, Unity, MongoDB, OpenCV*

[Repo](#) 2026

- Developed a surgical-training prototype using a **USB webcam and 3D-printed tool**; pitched it to Hack the North judges as a low-cost approach to hands-on practice.
- Mapped OpenCV tool poses into Unity coordinates to track instrument movement and integrated **ESP32 force readings** to drive simulated contact.
- Built a FastAPI and MongoDB backend to persist session telemetry and power replay and scoring, providing feedback on tool-path accuracy and pressure control.

TECHNICAL SKILLS

Languages: Go, Python, Java, SQL, C#, HTML/CSS, Bash

Databases: PostgreSQL, SQLAlchemy, SQLite

Developer Tools: Git, GitHub, Zed, Cursor, Codex, Docker, Postman, DataGrip, Unity

Cloud and Systems: Unix/Linux, Oracle Cloud, REST APIs, Webhooks, CI/CD